

Standard Form & Indices

Test 3 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 2^9 (1)

A2. 6^3 (1)

A3. 4^6 (1)

A4. $5^1 \times 5^2 = 5^3$ (1) = **125** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 3^2 = 1 \div 9$ (1) = **1/9** (1)

B3. $1 \div 10^5 = 1 \div 100000$ (1) = **0.00001** (1)

Section C — Writing standard form

C1. 8.3 (1) = **8.3×10^4** (1)

C2. 9.1 (1) = **9.1×10^6** (1)

C3. 2.5 (1) = **2.5×10^{-3}** (1)

C4. 6 (1) = **6×10^{-5}** (1)

Section D — Converting back, and calculating

D1. **5,100** (1)

D2. **0.0006** (1)

D3. $2 \times 4 = 8$, $10^3 \times 10^4 = 10^7$ (1) = 8×10^7 (1) already in standard form: **8×10^7** (1)

D4. $6 \div 3 = 2$, $10^6 \div 10^2 = 10^4$ (1) = 2×10^4 (1) already in standard form: **2×10^4** (1)

Section E — Powers to know, and estimating roots

E1. **128** (1)

E2. $10^2=100$, $11^2=121$ = between **10 and 11** (1)

Section F — Laws of indices with algebra

F1. **m^9** (1)

F2. **n^4** (1)

F3. **k^8** (1)

F4. $3 \times 5 = 15$ (1) $x^{2+4} = x^6$ (1) = **$15x^6$**

F5. $2^3 = 8$ (1) $y^{4 \times 3} = y^{12}$ (1) = **$8y^{12}$**

Section G — Fractional indices (Higher)

G1. $\sqrt{25} = \mathbf{5}$ (1)

G2. cube root of 64 = $\mathbf{4}$ (1)

G3. 5th root of 32 = 2 (1) $2^2 = \mathbf{4}$ (1)

G4. $\sqrt{4} = 2$ (1) $2^3 = 8$, $1 \div 8 = \mathbf{1/8}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.